

KRYLO — POC Architecture & Deployment

Status:  Live ·  Acceptance #3 PASSED

Last Updated: 2026-01-20

1. Purpose (POC Scope)

This Proof-of-Concept validates the core system behavior:

signal → aggregation → evaluation → expiry (reversion)

The goal is **mechanical proof**, not persistence, scale, or polish.

2. High-Level Architecture

text

[Client / Observer]

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[FastAPI + Uvicorn]

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[In-Memory Aggregation + Expiry Loop]

- **Signals:** Explicitly ingested via API
- **Aggregation:** In-memory, deterministic
- **Expiry:** Time-window based background loop
- **Persistence:** None (intentional for POC)

3. Hosting & Runtime (LOCKED)

Provider: Render

Service Type: Web Service

Tier: Free (short-lived acceptable)

Runtime Stack:

- Python 3
- FastAPI
- Uvicorn (ASGI server)

Cold starts reset memory. This is expected and acceptable.

4. Repository Structure (LOCKED)

Repo: ccmilme1-lang/krylo-soc

Branch: site

text

krylo-poc-backend/

|—— api.py

|—— requirements.txt

 **Critical:** Render Root Directory is set to krylo-poc-backend

This avoids Poetry auto-detection and monorepo ambiguity.

5. Render Configuration (CANONICAL)

Root Directory

text

krylo-poc-backend

Build Command

bash

pip install -r requirements.txt

Start Command

bash

uvicorn api:app --host 0.0.0.0 --port \$PORT

Why this matters

- python api.py **✗** caused early exit
- ASGI apps must be started via uvicorn

6. Dependency Management (LOCKED)

requirements.txt

text

fastapi

uvicorn

Captured lesson

- Render aggressively auto-detects Poetry
- Explicit dependencies + explicit root directory are required

7. Application Lifecycle (LOCKED)

- Uses **FastAPI lifespan** (@asynccontextmanager)
- No deprecated @app.on_event
- Background expiry loop starts safely at startup
- Clean shutdown semantics

8. API Surface (AS-BUILT)

text

GET /health

POST /signals/ingest

GET /signals/debug

GET /signals/live

GET /trust/score

Notes

- /health → service liveness
- /signals/ingest → signal entry point
- /signals/debug → observability for acceptance testing

9. Acceptance Status

Acceptance #3 — Aggregation Loop

 PASSED

Verified behaviors:

- Signal ingestion
- Deterministic aggregation
- Time-based expiry
- Clean reversion to baseline

10. Known Constraints (Intentional)

- In-memory only
- No persistence
- No auth
- Free tier cold starts
- Debug endpoints exposed

All are acceptable for POC.

11. Next Phases (Not Yet Implemented)

- Deterministic state machine (CALM / WATCH / ALERT)
- Multi-signal convergence
- AR HUD visualization layer
- Persistence + scaling (post-POC)

This document is canonical for the POC architecture.